

The Hyper-X Launch Vehicle: Challenges and Design Considerations for Hypersonic Flight Testing

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Abstract

The X-43A Program (Hyper-X) utilized a derivative of the Pegasus Space Launch Vehicle for atmospheric testing of the X-43A SCRAMJET Research Vehicle at Mach 7 and 10. NASA and Orbital Sciences Corporation developed the Hyper-X Launch Vehicle (HXLV) to meet the challenge of boosting the 2,800 lb X-43A to various test conditions in the atmosphere. This paper will address the design challenges and modifications required to standard Pegasus launch vehicle systems in order to survive the extreme aerothermal environments and successfully deliver the X-43A to multiple test conditions. The unique aerodynamic shape of the HXLV with the unshrouded X-43A attached, coupled with the high angle-of-attack flight profile, presented significant challenges to aerodynamics and Guidance and Control disciplines. Through lessons learned in the first flight failure and subsequent design modifications, extraordinary test condition accuracy on the second (Mach 7) and third (Mach 10) X-43A flights was achieved and will be discussed. Lessons learned on the HXLV can be applied to many design problems for Hypersonic Vehicles and Space Planes.

Nomenclature

α	=	angle of attack
β	=	angle of sideslip
γ	=	flight path angle
ρ	=	air density
σ	=	standard deviation
Q_{bar}	=	dynamic pressure
a	=	speed of sound
C_a	=	axial force coefficient
t	=	time
M	=	Mach
psf	=	pounds per ft ²
δ_e	=	elevon deflection

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I. Introduction

Hypersonic flight testing of advanced aerospace vehicles has long been an expensive, high-risk, and challenging endeavor. From 1959-1968 the X-15 program conducted 199 successful flights with three aircraft, 110 exceeding the Mach 5.0 boundary and the maximum achieving Mach 6.70 with USAF pilot William Knight on October 3, 1967. The Mach 6.7 flight was hardly uneventful, as shock waves burned through the leading edge of the modified lower ventral fin, fires were started in the engine bay, and pyrotechnic charges cooked off and ejected a dummy ramjet payload¹. From the last flight of the X-15 on October 24, 1968 to the first successful flight of the X-43A on March 27, 2004, atmospheric flight research of hypersonic vehicles has been limited to the Space Shuttle and ballistic missile reentry vehicles.

Several attempts were made to develop and flight test air-breathing hypersonic vehicles in the U.S. during the interim period between the X-15 and X-43A, however none were realized until the X-43A program. Descendants of the National Aerospace Plane (NASP) program attempted to flight test an airframe-integrated scramjet aircraft in the early 1990s utilizing Ballistic Missile assets, however each program was abandoned due to cost growth and schedule delays brought on by, among other things, significant development issues with the launch vehicle selected for the flight test. Other subscale flight tests, employing modified maneuvering reentry vehicles such as SWERVE on various launch vehicles, were proposed to test inlet performance, boundary layer transition, and advanced thermal protection concepts over the last two decades². These were not funded and were abandoned. Sounding rocket and surface-to-air missile experiments³ in the last decade provided data on scramjet performance in the atmosphere, however X-43A was the first successful experiment that achieved powered flight of an airframe-integrated scramjet propelled aircraft at hypersonic speeds. This achievement was made possible in part due to the maturity of the launch vehicle selected for the flight and its adaptability for the X-43A flight trajectory. The details of the launch vehicle selection process and trade study are presented below.

The Hyper-X (X-43A) Program is described in many references⁴. The X-43A Research Vehicle (RV) is 12 feet long, is 62 inches from wing tip to wing tip, and weighs approximately 2,800 lbm. The adapter designed and built by ATK-GASL to provide an interface and transition between the Pegasus booster and the RV weighed approximately 2,200 lbm for the Mach 7 configuration. Three flight tests were planned after a program restructuring in 1997, two at Mach 7 and one at Mach 10. Both test conditions were to be at a dynamic pressure (Q_{bar}) of approximately 1,000 psf.

II. The Hyper-X Launch Vehicle

Figure 1 illustrates the modification of the Pegasus Space Launch Vehicle (SLV) to the current configuration of the Hyper-X Launch Vehicle (HXLV). The vehicle design is the result of numerous trade studies and refinements as a result of those studies. The primary design constraints on the HXLV were to design a vehicle that achieves a wide range of test Mach numbers (5 – 10) and conditions (dynamic pressure, flight path angle) with a single booster configuration, and to maintain compatibility with NASA's B-52B-008 aircraft and the X-15 pylon. This resulted in minimum program developmental costs due to continuity of aerodynamics, systems, and interfaces between the launch vehicle and the X-43A RV on each of the different flight tests, and between the launch vehicle and the B-52 carrier aircraft. The baseline vehicle consists of an Alliant Techsystems (ATK) Orion 50S rocket motor, an aft skirt assembly which houses the fin/fin actuator assemblies, a Scaled Composites delta wing, a wing/fuselage fillet, and a Ballast/Avionics Module (B/AM) which houses the majority of the flight control electronics. The key differences between Pegasus and the HXLV are a stainless steel (Mach 7) or Aluminum (Mach 10) B/AM rather than a composite avionics deck; a HXLV-unique Thermal Protection System (TPS) that is designed to keep the composite structures at 90 °F during boost flight to Mach 10; an avionics architecture that allows for increased power switching, ordnance drivers, and telemetry capabilities; an upgraded Parker/Servo Controls Fin Actuator System (FAS), plus a reinforced fillet structure that allowed for increased aerodynamic loads associated with the HXLV trajectory.

The Aft skirt assembly is comprised of the aft skirt structure, three composite double-wedge fins, and the FAS. The aft skirt structure remained largely unchanged from the baseline Pegasus design for the Mach 10 flight, however, CG considerations for the second Mach 7 flight drove the design team to a steel structure built to accommodate increased aft ballast. The aft skirt is a basic all aluminum structure of conventional ring and stressed skin design. It is constructed in two halves to allow installation over the larger diameter motor nozzle. Each Scaled Composites fin is constructed of identical one-piece solid foam core and wet-laid graphite composite construction with a central tapered polygon titanium shaft. Thermal protection for the low-temperature composite fins was a challenge for the Mach 7 and 10 flights due to extreme heating in the interference region between fin root and

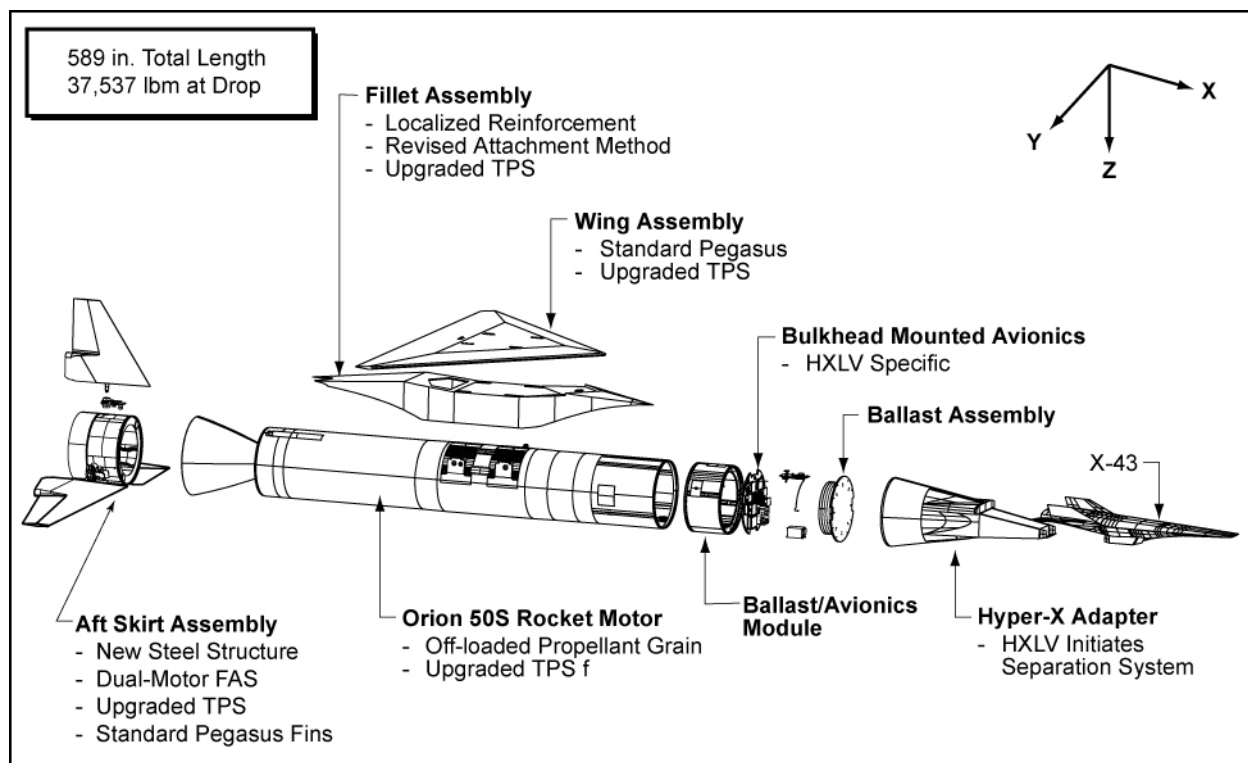


Figure 1. The Hyper-X Launch Vehicle.

fuselage, and the limited allowable thickness available. Silica phenolic tiles were fabricated and bonded for the fin leading edges, and syntactic foam was either molded or applied in sheets to various regions of the fin surface. The fin shaft is press fit into the fin actuator assembly consisting of two permanent magnet motors, high-strength five stage reduction gear train, position potentiometer, and a machined steel housing. The aft skirt houses the actuator FAS Electronics Control Unit (ECU) which contains the signal conversion and power output boards that will accept commands from the flight computer and convert them to analog signals to actuate the fins over a substantial range of motion. This ECU was redesigned following the first flight to accommodate the dual-motor actuators and increase the output torque to almost double the baseline Pegasus system. Additional thermal batteries were utilized for the Hyper-X second and third flights which are housed on an auxiliary battery unit mounted to an aft skirt access panel. The aft skirt assembly also contains thermal and strain gauges mounted internally that were monitored by the flight encoder housed in the B/AM.

The delta wing assembly is based on a curved, double spar surrounding a central box structure with machined aluminum fittings at three of the four corners. The forward port fitting is a stainless steel machined fitting that is designed to react all vehicle lateral loads during taxi, take-off, captive carry, and landing. Captive flight loads from the carrier aircraft hooks as well as launch aerodynamic loads from the wing are transmitted to the motor saddle via a steel tube truss that bolts to the lower faces of these fittings. The wing panels and wing box structure contain embedded thermal and strain gauges that were monitored during all phases of flight by the encoder in the B/AM. The truncated delta planform wing has an aspect ratio of 4 and an 8 percent double wedge profile truncated to 20 cm thickness. The leading edge is blunt to limit aerothermal heat transfer, while the trailing edge is aerodynamically sharp for Reynolds numbers encountered in flight. The wing TPS consists of molded leading edge tiles, either silica phenolic (Mach 7) or carbon phenolic (Mach 10). These tiles were designed to withstand the severe stagnation point heating on the leading edge, as well as shock-enhanced heating from the vehicle bow shock and RV control surface shocks. Acreage TPS consists of cork, RTV, or cork/RTV hybrid sheets bonded to the composite structure. This is a significant upgrade from Pegasus, which utilizes flexible silicon-based (RTV) sheets bonded to the leading edges and cork elsewhere.

The single stage rocket motor is a Hyper-X specific booster based on the qualified Alliant Techsystems Orion 50S booster. The propellant grain was modified slightly from the baseline for the fully-loaded (Mach 10) configuration, and modified extensively for the second flight at Mach 7 by machining over 3,300 pounds of propellant out of the cast motor. The booster was designed specifically for air launch applications, which includes

qualification for firing following cold soak at altitude. The design features a carbon-phenolic nozzle with a 3D carbon-carbon integral throat entry. The motor case is filament wound and co-cured with internal insulators using Alliant patented techniques. Wound into the motor case is the aluminum wing saddle which reacts the captive carry and aerodynamic wing loading through the motor case. The motor case, like the other components contains embedded thermal and strain gauges that are monitored constantly by the flight encoder.

The wing-fuselage fillet structure, manufactured by Scaled Composites, is made of low stress Nomex foam with graphite facing sandwich construction. The only loading of the fillet is due to aerodynamics, and since the flight profile of the Hyper-X mission is significantly different than a standard Pegasus considerable effort was put into analyzing the load cases. Higher sideslip loads led to an increased ply layout in the forward nose and midsections of the fillet. The increased loading has also led to a modification of the fillet installation process. The standard Pegasus method of attachment at the forward nose uses an inverted composite T-Bracket which is bonded to both the motor case and the fillet. The HXLV method of attachment used a G-10 shoe which will be bolted through the motor case and the fillet nose. A sliding joint is used between the fillet and the wing to accommodate deformation in the wing shape during the maximum aerodynamic load condition, and the fillet design also accommodates motor case expansion during thrust flight. The interface between the fillet and motor case is protected by a 4 inch silicone joint, which is bonded to the fillet and motor case and provides a flexible aerodynamic seal around the perimeter of the fillet.

The B/AM is a new design specific to the Hyper-X program. The B/AM serves as the mounting location for the majority of the flight avionics, including the flight control, flight termination, ordnance initiation, and power subsystems. These avionics are mounted on a composite structure forward of Stage 3 on the baseline 3-stage Pegasus. The HXLV components, with the exception of the inertial measurement system, are mounted on an shock isolated, aluminum skinned honeycomb bulkhead. The Inertial Navigation System (INS) is installed in the B/AM on a longeron mounted plate. The B/AM also serves as the trim mechanism for final vehicle mass, CG, and pitch moment of inertia through the use of ballast plates and can accommodate up to 5,000 pounds of ballast to support multiple test conditions. The B/AM is the interface between the Hyper-X RV Adapter and the launch vehicle which allows for electrical and fluid interfaces between the two. Due to a requirement to maximize vehicle stiffness for the Mach 7 flights, the Mach 7 B/AM was designed using 15-5PH CRES rolled ring and roll formed skins plus 304L CRES square tubing longerons to increase stiffness. The B/AM contains thermal, strain, and acceleration transducers that are monitored by the flight encoder.

The avionics suite employed on the HXLV consists of flight proven hardware from many Orbital programs. The principle navigator is the Litton LN-100LG GPS-Aided Inertial Navigation System. The unit provides three types of output, inertial only, GPS only, and a merged solution via a SDLC link to the on board SBS-or flight computer. The unit is flown in a GPS-aided mode during captive carry on the B-52 by using a B-52 GPS antenna, and in free-inertial mode following drop due to the limited time of flight during boost. The flight computer manufactured for Orbital by OR Industrial Computers, is a 68030 based computer which employs two 68302 based communication boards. The flight computer will use the inertial only measurements to generate the path steering guidance commands necessary to control the three fin actuators in the aft skirt assembly. Support avionics for power switching/routing and ordnance initiation is accomplished through the use of Orbital designed and built electronics. All on board power is provided by two separate 5.0 A-Hr NiCad battery packs. The ordnance subsystem is a fully redundant subsystem featuring independent batteries. Flight termination, in the event of a boost vehicle failure, is accomplished via a range qualified fully redundant system. This system was functioned successfully on the errant first flight, and then functioned deliberately following X-43A separation on Flight 2 (F2) and Flight 3 (F3) to expend the ordnance. The use of the range-certified Pegasus Flight Termination System (FTS) was a major cost savings and risk reduction factor in using an existing booster. Each side of the FTS features a dedicated battery, a UHF antenna, coupler, and receiver, an Orbital built Flight Termination Logic Unit (FTLU) and an electrical to explosive interface device. All data collection, conversion, and transmission is handled by a L3 Conic PCM-600 flight encoder.

III. Design Trades Leading to HXLV Selection

The concept of using Pegasus for hypersonic flight testing had been recognized by the community for some time leading up to its selection for Hyper-X in 1996, as described in Reference 1. Pegasus is ideally suited to the Hypersonic Flight Test problem due to its Stage 1 control system, which utilizes three single-wedge fins for aerodynamic control of the Stage 1 motor, which has a fixed nozzle. Since the X-43A flight test profile required separation in the atmosphere, aerodynamic control was required following motor burn-out. The air-launch aspect of Pegasus provides lower boost-phase structural and thermal loading, and additional flexibility for launch location, which is a more significant issue for Hypersonic Flight Testing than for traditional launch vehicle missions as

discussed below. Finally, the substantial performance of Pegasus provides the flexibility for lower-cost solutions to hypersonic flight test issues. Other boosters considered did not have aerodynamic control during or after booster burn and were performance limited, which would have resulted in a development program on the booster. In the end, NASA selected the Pegasus derivative system due to its unique ability to boost the Hyper-X Research Vehicle (HXRv) to the three baseline test conditions in a single configuration, which reduced the risk to the program by eliminating the need for multiple launch vehicle configurations and for substantial development efforts on other booster systems.

The original requirements for the Hyper-X program included four flights at three separate Mach numbers: one at Mach 5, two at Mach 7, and one at Mach 10. Separation condition requirements were similar for all three flights, with a separation dynamic pressure (Q_{bar}) of approximately 1,000 psf and a flight path angle (γ) of +2 degrees. The Mach 5 flight was deleted in December of 1997 shortly after CDR due to budget constraints and a ground test approach that satisfied NASA's research requirements. Trade studies in 1996 conducted by various Government and industry teams examined several different booster options. The primary candidates studied by Orbital were the air-launched Pegasus derivative, a ground-launched ATK-Thiokol Castor IVB configuration for the Mach 5 and 7 flights, a ground-launched Taurus Stage 2 (Orion 50S GL) for the three test conditions, and an ATK GEM-VN motor that had slightly better performance than the Castor IVB, but still did not achieve the Mach 10 test condition. For all but the Pegasus Stage 1 approach, the addition of a post-boost aerodynamic control system would be required. Pegasus was selected based on favorable trade study results in the following categories:

- 1) Low risk due to the fact that a single configuration could be developed to meet all three flight test requirements
- 2) Operational advantages to air launch
- 3) Availability of the NASA B-52B aircraft for the program, which had been used for the first six Pegasus Flights
- 4) Lower integrated heating during boost for air launch
- 5) Existing aerodynamic control system
- 6) Flight-proven system

Table 1 provides a summary of the early trade study results based on 3DOF analysis with a 2200 lbm Research Vehicle and 500 lbm Research Vehicle adapter.

Table 1. Early Trade Studies for Hyper-X Missions.

Configuration	Mission	Margin (Ballast - lbm)	Peak Heating (normalized*)	Integrated Heat Load (normalized*)	Control System
B-52 Launched Orion 50S	M5	20,000 (a)	0.21	0.26	All Aerodynamic; Upgraded Pegasus FAS
	M7	10,600 (a)	0.44	0.39	Same
	M10	2,000 (a)	1.0	1.0	Same
Ground-Launched Orion 50S	M5	11,000	0.20	0.34	TVC with Fins and Ailerons – New Development
	M7	5,350	0.42	0.46	
	M10	500	1.0	1.54	
Ground-Launched Castor IVB	M5	5,800	0.20	.015	TVC with 4 Fins and Ailerons – Upgrade of Existing Orbital System
	M7	1,400	0.423	0.29	
	M10	Mach 8 Maximum	N/A	N/A	
Ground-Launched GEM-VN	M5	10,000	0.25	0.29	TVC with Fins and Ailerons – New Development
	M7	4,000	0.47	0.29	
	M10	0 (Mach 9.8)	N/A	N/A	

*Normalized to Orion 50S Air-Launched Mach 10 Trajectory

(a) 40,000 ft drop Altitude

Appropriate trajectory constraints for each vehicle were applied based on Orbital's design experience with these vehicles, specifically structural (α - Q_{bar}) limits, control system limits (TVC or FAS), acceleration limits of the rocket motors, α limits, etc. The margin numbers referenced are based on preliminary, low-fidelity 3-DOF POST runs using mass and aerodynamic properties typical for Orbital's fleet of Suborbital launch vehicles. No margin is included on top of the value provided. For the GEM-VN, for example, with no ballast the vehicle could still not achieve the Mach 10 condition. Heat loads were computed for a reference sphere and compared on the different trajectories. A comparison of the relative integrated heat load between the ground-launched and air-launched Orion 50S reveals the penalty for ground launch to Mach 10, with ground launch having greater than 50% higher heating than air launch for the same motor. Although the Castor IVB provided a flight-proven (Orbital Vehicles, Maxus Sounding Rocket) solution for the Mach 5 and 7 flights, NASA elected to pursue a single configuration for all three missions and selected Pegasus. At the time, recovery of the X-43A Research Vehicle either in a runway landing or via a water recovery system was part of the baseline, making the air-launch feature particularly attractive for use with existing test ranges. The GEM-VN vehicle had not been flown other than as a strap-on booster for Delta II, so this approach was deemed high risk by NASA.

For the air launched option, note that a 40,000 ft, Mach 0.8 drop condition was assumed. With 10,000 or 20,000 lbm of ballast, the store weight of the HXLV "stack," including the adapter and Research Vehicle, would exceed the 41,400 lbm capability of NASA's B-52B 008 with the X-15 pylon and Pegasus adapter. In addition, a coast period to burn off excess performance following booster burn-out was ruled out due the severe thermal loading this would place on the LV and RV. This design problem applied to both the Mach 5 and 7 flights, and the design team traded different approaches to the problem to keep the total vehicle weight and CG within the B-52 pylon constraints. The three approaches to this energy management problem were (1) Propellant offload to reduce the available total impulse and thus the ballast requirement, (2) perform energy management maneuvers during boost to reduce the separation velocity, and (3) lower the drop altitude and decrease the drop Mach number to increase the energy requirements for a given test condition. Structural and control system limitations of the baseline Pegasus systems quickly eliminated the second option due to the strong desire to minimize changes to the launch vehicle. For the Mach 5 flight, an unrealistically low drop altitude was required (<10,000 ft) in order to achieve the test conditions without energy management, leaving propellant offload as the only option for this flight.

For the Mach 5 vehicle design, the requirement to offload propellant from the Orion 50S generated another design activity. Since the motors were not yet fabricated, ATK identified an approach to reducing the propellant load by up to 7,000 lbm that employed a simple thickness increase to the insulator to displace the desired mass of propellant. Thus, the standard casting tooling for the Orion 50S could be utilized, no propellant machining operation was required, and the ballistics at ignition and during the majority of the burn would be identical to the baseline system. ATK demonstrated fabrication techniques for this thicker insulator prior to the deletion of the Mach 5 flight in December, 1997, and this remains a viable approach for future missions.

For Mach 7, the trade was less clear. During the early design phase of the program, the design team evaluated a fully loaded Orion 50S dropped at low altitude (21,000 ft) and low drop mach vs. an offloaded motor at higher altitudes. A significant offload was required for the high altitude drop, however the design and development effort for the Mach 5 flight had identified a feasible and moderate-risk approach to offloading a significant mass of propellant. In assessing overall mission risk, however, the team concluded a first flight of the offloaded motor on the first Hyper-X launch was layering excessive risk, and elected to pursue the lower altitude, heavy booster approach. This decision would prove to be a contributor to the Flight 1 (F1) failure, as the first 20 seconds of flight with this approach resulted in approximately double the transonic Q_{bar} compared with Pegasus--and minimal margin for error.

Prior to contract award to Orbital in February, 1997, structural modeling activities revealed that the first longitudinal bending mode of the vehicle was well below requirements for the launch vehicle control system. Since most of the compliance in the system was in the adapter-to-free-flyer joint and the RV and adapter structures, these designs were re-evaluated. This discovery had a major program impact, as the Research Vehicle and adapter teams had to return to the conceptual structural design of the X-43A and the booster adapter and design for stiffness. Ultimately, the primary keel structure of the RV and the entire booster adapter were switched from lighter weight materials to steel, with a dramatic effect on total weight. Where pre-contract award mass properties for the payload (booster adapter plus RV) were approximately 3,300 lbm, the CDR mass properties for the Mach 7 RV plus adapter were 5,130 lbm. From Table 1, note that this mass growth would have eliminated the Castor IVB configuration for the Mach 7 flight (the Castor IVB has similar stiffness requirements). Thus the selection of the Pegasus derivative for the Mach 7 flight proved fortuitous due to the large amount of available mass margin. This weight growth also eliminated the margin for the Mach 10 flight, so the team pursued designs for this mission that reduced the weight of the B/AM and booster adapter. The Mach 10 B/AM was constructed of Aluminum, and the booster adapter was redesigned to eliminate unneeded subsystems and add some aluminum components.

IV. Flight Results

Previous authors have described the Hyper-X F1 results⁵, so only a summary is presented here. The F1 HXLV was configured for Mach 7 flight, using the baseline HXLV avionics, a fully loaded Orion 50S rocket motor, a steel B/AM with approximately 3500 pounds of ballast, aluminum aft skirt with standard a standard Pegasus FAS, and a Mach 7 Thermal Protection System. The planned F1 Trajectory is shown in Fig. 2. The final trajectory release utilized a drop condition at approximately 25,000 feet and $M=0.5$, pulled up to a maximum alpha of 13.1 degrees, and pushed over to a maximum negative alpha of -4.4 degrees to level the trajectory at the separation condition of $Q_{bar} = 1029$ psf and Mach 7.05. Comparison of the dynamic pressure with a nominal Pegasus boost is shown in Fig. 3, illustrating the much higher transonic Q_{bar} associated with this trajectory due to the low altitude drop.

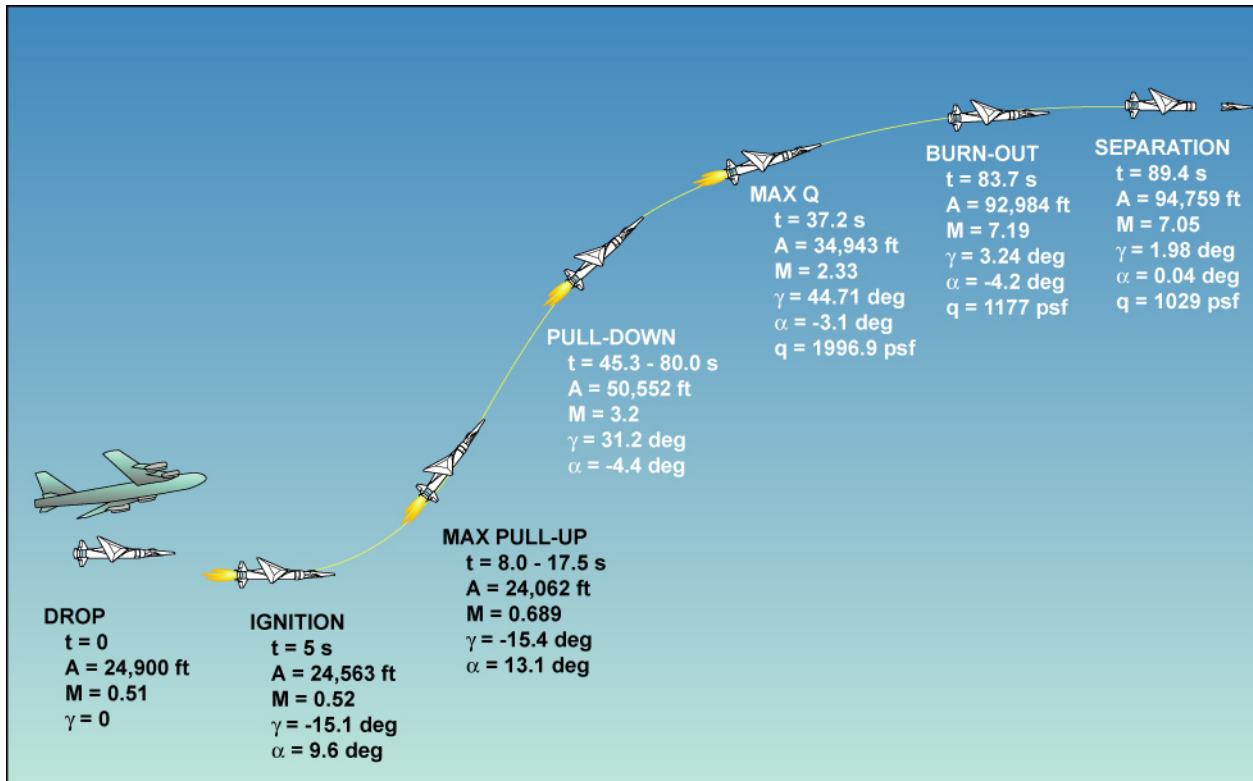


Figure 2. Nominal Flight 1 Trajectory Profile.

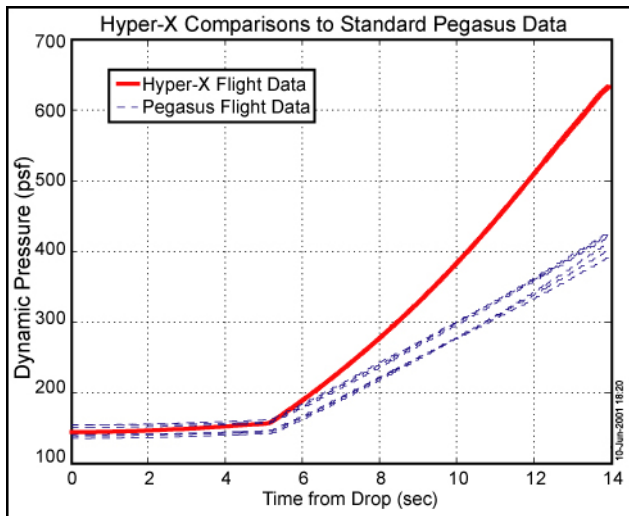


Figure 3. Hyper-X Transonic Dynamic Pressure Comparison with Pegasus Standard.

F1 launch occurred on June 2, 2001. Pre-launch testing and flyout to the test point were nominal, and winds and atmospheric conditions were benign. The booster stack successfully separated from the B-52 carrier aircraft, motor ignition occurred as designed at $t+5$ seconds, and the control system established the pull-up alpha of approximately 13 degrees per plan. At approximately 11 seconds and Mach 0.8, a divergent roll oscillation began that grew in amplitude until first the rudder, then the starboard fin actuator stalled. By 13 seconds, the vehicle had exceeded the design envelope, backdriving the starboard fin actuator until the starboard fin failed and departed the vehicle. Following this structural failure, which came at expected failure loads based on trajectory reconstruction, the vehicle quickly tumbled out of control followed by multiple structural failures and the loss of the mission. The Range Safety Officer (RSO) sent the Flight Termination commands that

successfully destroyed the booster prior to safety boundaries being exceeded. Figure 4 provides an example of the flight data in rudder and elevon position, and roll rate illustrates the vehicle dynamics prior to fin failure.

NASA immediately convened a Mishap Investigation Board (MIB) that investigated the failure for the next six months in an intensive data analysis, modeling, and test effort that provided a solid starting point for return to flight⁶. The MIB released its report in May, 2002. That the control system was designed with too much gain was immediately apparent; what took the next six months was determining the sources for the improper gain values in the autopilot design. The MIB found that three primary factors contributed to the failure: inaccurate modeling of the Fin/FAS system (primarily unmodeled compliance outside the servo loop), aerodynamic model deficiencies, and mass properties modeling inaccuracies. The first and second factors were the primary drivers, however several additional factors contributed to the failure. In conducting the failure investigation, which examined the overall Launch Vehicle design in great detail, the vast majority of the designs were re-validated. However, it became increasingly clear that utilizing the existing Pegasus design within the constraints imposed by the test conditions and B-52 pylon limits was extremely high risk—the envelope was too narrow for a first flight of such a complex airframe.

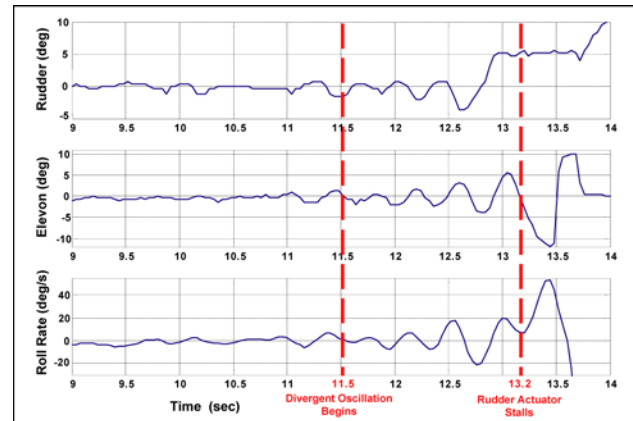


Figure 4. Flight 1 Roll Rate and Elevon Position.

A. Return To Flight (RTF)

Prior to completion of the MIB in the winter of 2001/2002, the team embarked on several return to flight trade studies in order to complete the second Mach 7 and final Mach 10 missions. The team approached the RTF activity with the following goals: 1) Increase control system power margin to achieve a minimum reserve of 50 percent under worst case predicted flight conditions. 2) Comply with revised B-52/X-15 pylon loading constraints. 3) Increase fidelity of nominal and dispersed design models in order to accurately assess system margins prior to flight. Changes were to be implemented only where significant margin improvements (flight envelope expansion) and robustness could be achieved within programmatic constraints (cost and schedule). The second goal listed was the result of NASA revisiting the design, analysis, and historical performance data of the X-15 pylon and Pegasus adapter. The conclusion of this study imposed a significantly lower allowable drop weight for the stack and more restrictive CG location than that used for F1 mission design. This result had a significant impact on several LV trade studies and associated redesign decisions for both the M7 and M10 missions.

Based on these goals, a number of trade studies were undertaken to identify logical changes to the trajectory and launch vehicle control system design. The primary goal to increase the flight envelope and associated control system power margins was addressed on three fronts: 1) Trajectory trade study that placed high emphasis on reducing control surface loading as much as possible, particularly in the subsonic/transonic regime of flight. 2) Control system hardware trade study that considered modifications to the existing aerodynamic fin control system or wholesale replacement of this system to increase control power. 3) Autopilot design trade study to determine the optimal architecture for achieving tight trajectory control under dispersed vehicle aerodynamic and trajectory conditions as well as having robust disturbance rejection properties. Satisfaction of revised X-15 pylon limitations and significantly improving the fidelity of design models was accomplished in parallel with these trade study efforts.

Control system modifications were examined to determine if a more robust design was possible within other program constraints. These were in four categories: the addition of Thrust Vector Control (TVC) in place of the fixed Orion 50S nozzle; redesign of the aerodynamic control surfaces; FAS upgrades – either replacement of the baseline Pegasus system or upgrades to it; and autopilot modifications or a departure from the baseline Linear Quadratic Regulator (LQR) approach. The details of these studies are beyond the scope of this paper, however the result was a decision to upgrade the existing FAS to significantly increase the output torque and improve the control power margins. Additional studies included a liquid ballast transfer study which addressed landing load limitations of the B-52, and a L-1011 study which examined the feasibility of using Orbital's L-1011 Pegasus Carrier Aircraft for Hyper-X due to its greater lift capability than the B-52.

The trajectory trade study placed high emphasis on reducing control surface loading as much as possible, particularly in the subsonic/transonic regime of flight. The goal was set at achieving a transonic dynamic pressure of 320 psf which is $\frac{1}{2}$ of the F1 trajectory (this value is also comparable to typical Pegasus missions). In addition, it

was deemed desirable to minimize time in subsonic/transonic flight as much as possible so the drop speed was set to the maximum B-52 capability of Mach = 0.8. Given this, the only meaningful way to achieve the transonic dynamic pressure goal was to raise the drop altitude to the maximum B-52 capability of 40,000 ft. Obviously, the majority of the Q_{bar} reduction results from the lower atmospheric density at higher altitudes. To a lesser extent, Q_{bar} reduction also occurs with increasing altitude due to the lower ambient air temperature which provides a lower true airspeed at Mach = 1. Given these decisions, the initial conditions at the start of the trajectory are essentially fixed.

The LV/RV separation requirements for the RV flight experiment are defined in terms of Mach, flight path angle (γ), and dynamic pressure (Q_{bar}). The Mach and flight path angle set the velocity vector requirement and the dynamic pressure essentially restricts the altitude at which separation can occur to a narrow range. Rearranging the definition for dynamic pressure illustrates this point.

$$Q_{\text{bar}} = 1/2 \rho (M * a)^2 \rightarrow \rho * a^2 = 2 Q_{\text{bar}} / M^2 = f(\text{alt}) \quad (1)$$

Given the initial and final energy states of the trajectory are essentially fixed, it is illustrative to look at LV performance in terms of velocity losses. The net ΔV capability of the vehicle during the boost phase can be expressed as:

$$\Delta V_{\text{net}} = V_{\text{sep}} - V_{\text{drop}} = \Delta V_{\text{ideal}} - \Delta V_{\text{atm}} - \Delta V_{\text{drag}} - \Delta V_{\text{grav}} \quad (2)$$

where	$\Delta V_{\text{ideal}} = \int (thrust/mass) dt$	Maximum amount of velocity change the motor can impart in a vacuum.
	$\Delta V_{\text{atm}} = \int (P_{\text{atm}} * A_{\text{exit}}/mass) dt$	Velocity loss due to motor operation in the atmosphere
	$\Delta V_{\text{drag}} = \int (drag/mass) dt$	Velocity loss due to vehicle drag
	$\Delta V_{\text{grav}} = \int (grav * \sin \gamma / mass) dt$	Velocity loss due to change in potential energy (i.e. change in altitude)
	$V_{\text{drop}} =$	velocity at drop at 40kft (M = 0.8)
	$V_{\text{sep}} =$	velocity at sep at 95kft (M=7.075)

The terms ΔV_{atm} and ΔV_{grav} are essentially functions of the change in altitude between drop and separation. We concluded earlier that these cannot be varied. In addition, it can be shown that these terms are small compared to either ΔV_{ideal} or ΔV_{drag} . Adjusting the drag term is only possible through either significant vehicle mold line changes (i.e. increase C_a) and/or increasing the integrated dynamic pressure profile of the trajectory. Vehicle mold line changes would invalidate the existing aerodynamic database and only be applicable to the Mach 7 mission while increases in integrated dynamic pressure would require significant redesign of both structural and thermal protection systems (which were already delivered and integrated) of both the LV and RV. For these reasons, it was quickly determined that this was not a viable method for adjusting vehicle performance within program cost and schedule constraints. This left only vehicle ballast and propellant off-load as viable methods for adjusting vehicle performance. In order to satisfy the X-15 pylon weight limitations for the Mach 7 mission it was clear that ballast alone would not provide adequate performance scrubbing and that propellant would have to be removed from the motor. 3-DOF trajectory simulations were conducted to determine that approximately 3300 lbs of propellant would have to be removed to meet all Mach 7 mission constraints. For the Mach 10 mission, the fully loaded motor in combination with no ballast still satisfied the revised X-15 pylon weight and CG limitations.

Since the F2 motor was already fabricated and integrated in the field, a machining operation was required to remove propellant instead of the approaches studied as part of the Mach 5 configuration. The approach to machining the propellant grain was considered iteratively with the Mach 7 trajectory design trades described above, and the offload amount of 3,345 pounds was selected to minimize risk for the machining operation, as well as for maintaining a viable ignition grain geometry and staying within the Orion 50S design constraints. This value is the total mass of the fin region of the propellant grain, and eliminated the fins from the motor. This had a significant effect on the motor thrust profile, and produced a pronounced tail-off at the region of maximum dynamic pressure in the trajectory, reducing loads on the vehicle and control system (Fig. 5). Since a static fire of this significant modification was not possible due to funding and schedule constraints on the program, NASA formed an independent review team that assessed all aspects of the design change. Extra attention was paid to ballistic modeling (no static firing), combustion instability, thermal margins due to the longer burn time, and igniter performance with the larger free volume at ignition. ATK developed the machining techniques to safely remove the propellant, completed all design analyses (structural, thermal, ignition, ballistics), demonstrated the machining

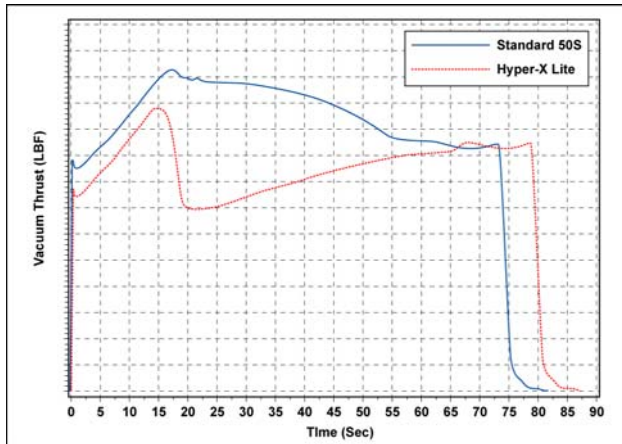


Figure 5. Offloaded Thrust Curve vs. Fully Loaded Motor.

of these design changes was to increase the output torque of the system by almost 90 percent above the Pegasus system. Detailed design studies were conducted of the upgraded FAS and new non-linear models of the system were developed. The team conducted development tests on the new system to validate these models, with particular focus on compliance of the entire aft skirt, actuator, and fin system. Qualification testing of the new actuator, electronics unit, and auxiliary thermal battery unit were completed successfully in the spring of 2003, nine months from the start of development. However, the first production FAS failed acceptance testing, revealing design issues with the electronic unit related to the high current switching required for increased performance. A Complex Programmable Logic Device (CPLD), utilized in the power control boards for each fin channel, was susceptible to CMOS latch-up caused by high noise environments present during high slew rate maneuvers, when the system would current limit due to start/stop/reverse inertial loads. Several design modifications were made to the ECU to mitigate this issue, and the changes were implemented, verified and the flight system delivered to the field for integration two weeks prior to the F2 launch.

A significant aerodynamic development effort⁷ was undertaken to improve the fidelity of the aerodynamic models, and an extensive test and modeling effort provided a new FAS model for 6-DOF and linear autopilot analysis. The aerodynamics of the HXLV airframe are extremely complex, particularly interaction of the RV-generated vortices with the HXLV rudder (Fig. 6). In the F1 failure, rudder actuator stall was not predicted at the flight condition where it occurred, and this was discovered to be due to the highly non-linear behavior of the

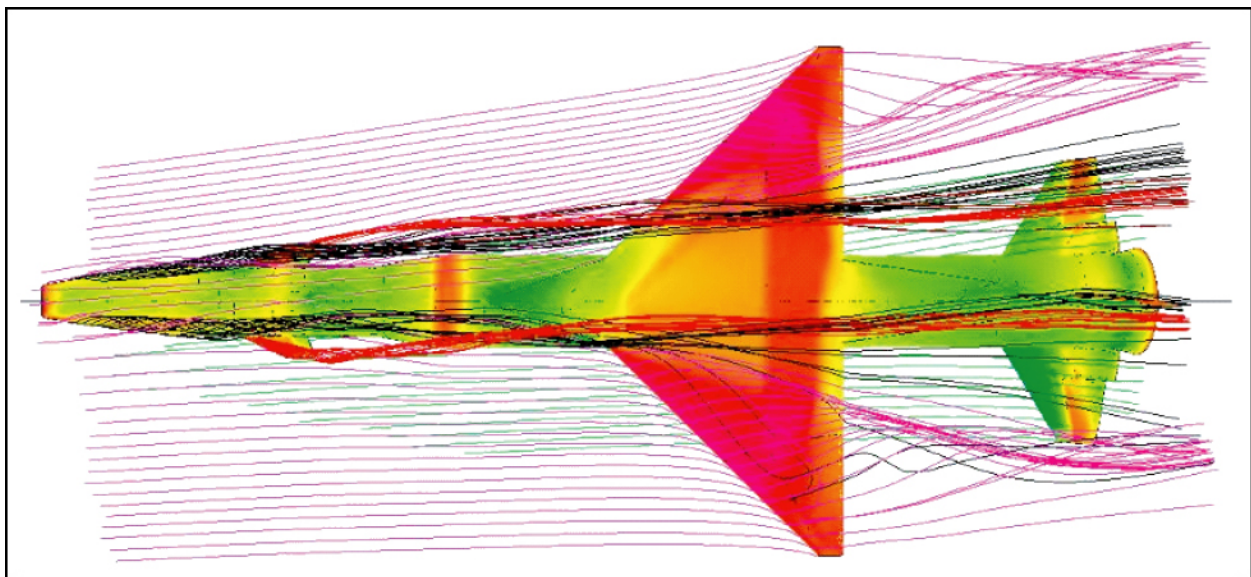


Figure 6. CFD Results at High β .

approach on an inert motor simulator, disassembly, machining, and reassembly of the flight motor in six months. The magnitude of the offload was one of the largest ever performed on a solid rocket motor.

Another major modification to the system was an upgrade to the baseline Pegasus FAS. Larger aerodynamic uncertainties utilized in RTF 6-DOF simulations and added conservatism for control margins dictated a significant upgrade in output torque. A survey did not reveal any existing system that met requirements, so the existing system was modified by upgrading the gear housing to steel from aluminum, replacing the baseline gear material with a more ductile alternative, adding a second motor in a torque summing arrangement, electrically in parallel with the first, and redesign the control electronics to accommodate the second motor and a higher current limit. The net result

rudder hinge moment resulting from this vortex interaction at high alpha and beta. This was not modeled pre-F1 due to the sparseness of the wind tunnel database at these flight conditions.

In summary, the F2 Vehicle and Mission Design differed from F1 in the following areas:

- Drop conditions changed to 40,000 ft, Mach 0.8 from 24,000 ft, Mach 0.5
- FAS upgraded to provide 90 percent more output torque
- 3345 pounds of propellant machined from Orion 50S motor
- Steel aft skirt replaced aluminum skirt to shift stack CG aft
- Overall stack weight decreased by 3863 lbm to 37,537 lbm

The nominal trajectory profile for F2 is provided in Fig. 7. The alpha profile for this mission is presented in Fig. 8.

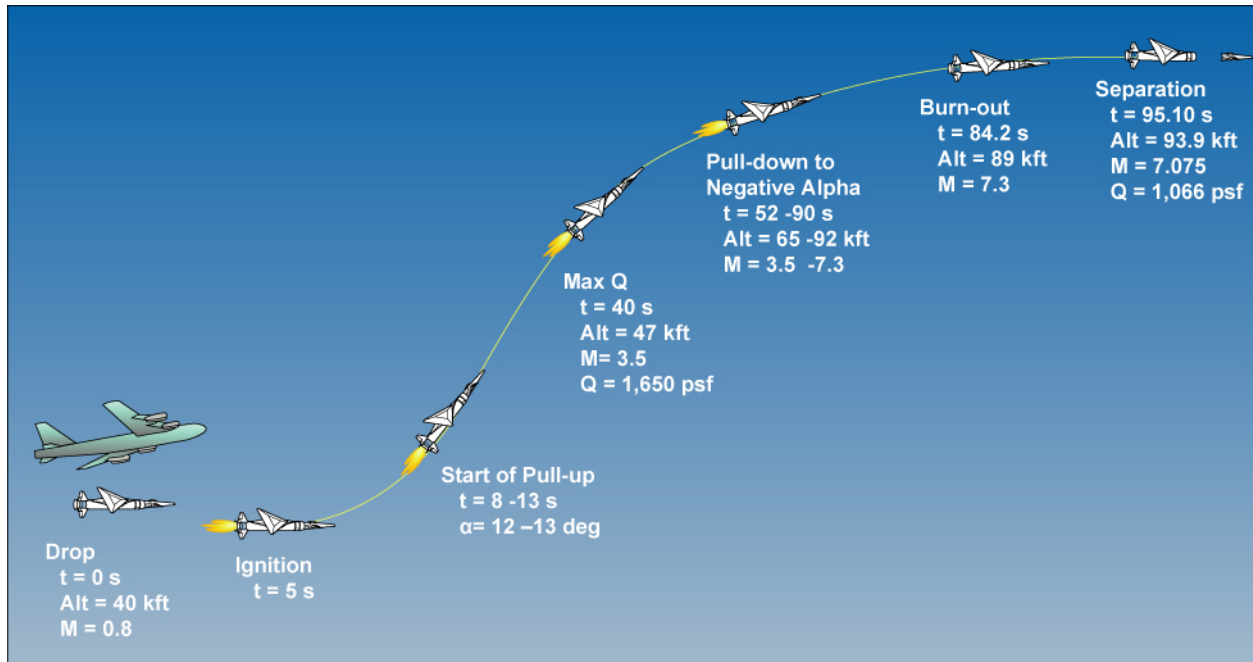


Figure 7. Nominal F2 Flight Profile.

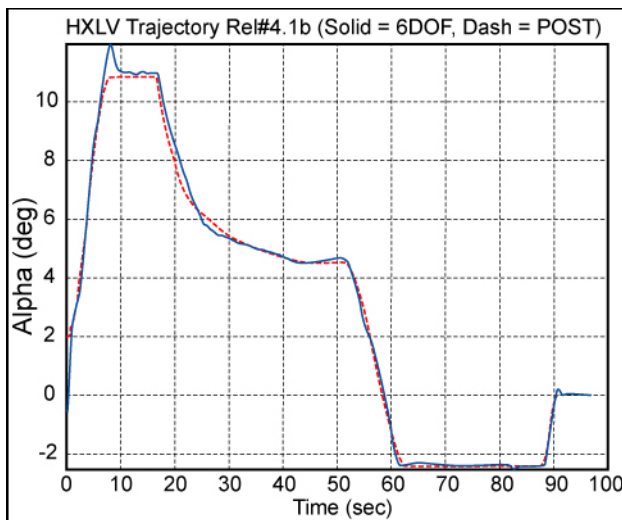


Figure 8. Nominal F2 Alpha Profile.

B. Flight 2 Results

The second flight of Hyper-X was conducted on March 27, 2004 (Fig. 9). Flight performance exceeded all expectations on this flight, primarily due to the limited amount of data collected on F1 (up to 13 seconds). Again, pre-flight testing and weather were nominal. The B-52 achieved the desired drop conditions within tolerance, and separation from the B-52 was successful. Roll rates at tip-off were higher than expected, but were successfully captured by the HXLV control system shortly after it was enabled at a safe distance from the carrier aircraft.

Throughout flight, control system performance was nominal. There was no evidence of the transonic instabilities observed on F1, indicating optimal gain selection and validating the new aerodynamic and control system models. As shown in Fig. 10, a wing rock phenomenon was observed at approximately



Figure 9. HXLV Ignition on Flight 2.

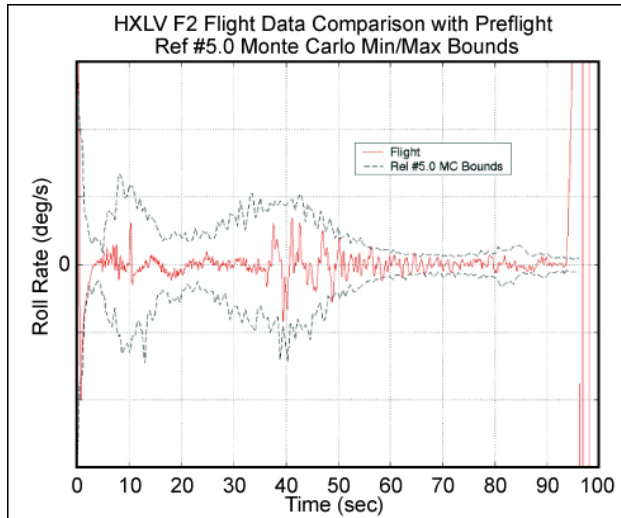


Figure 10. F2 Roll Rate Vs. Time.

Table 2. Flight 2 Separation Condition Performance.

Parameter	Target	Flight Data*
Separation Time	<100 sec	93.466 sec
Separation Alt	93,932 ft	94,069 ft
Mach	7.075	6.998
Flight path Angle	2.0 deg	2.42 deg
Dynamic Pressure	1,066 psf	1,037 psf
Alpha	0.0 deg	0.09 deg
Beta	0.0 deg	0.0 deg

*Derived from LV INS and 2000Z DOL San Nicholas Balloon Data

flight profile, keeping the core systems (notably the FAS) from the successful Mach 7 flight. With this decision and the addition of wind tunnel data specific to the Mach 10 TPS Outer Mold Line (OML), the final Mach 10 trajectory was developed three months after F2 as shown in Fig. 11. The additional wind tunnel data on the Mach 10 TPS OML proved valuable, as the relatively minor leading edge shape on the wing and fins had a significant effect on certain aerodynamic parameters, most notably fin hinge moments and asymmetric wing loading due to boundary layer transition effects. These findings are the subject of an upcoming paper by Luis Bermudez.

Significant changes to the Mach 10 TPS were required due to the higher heat loads on this trajectory. Design analyses were conducted in 2001 and 2002 by Orbital and ITT Aerotherm, Inc. to identify changes required from the Mach 7 TPS on all regions of the vehicle with the exception of the motor case, which implemented the Mach 10 TPS thickness for the Mach 7 flights as well. The primary areas of concern for the Mach 10 flight were the wing and

Mach 1.0, consistent with prior Pegasus flights and wind tunnel data. The addition of a roll disturbance model to the 6-DOF analysis for use in gain selection resulted in an autopilot design robust to this phenomenon, and it was quickly captured and damped by the control system. Activity from 35 to 50 seconds is well correlated with winds measured by balloon sondes at two-hour intervals leading up to the launch.

Overall performance of the launch vehicle was nominal up to separation. The HXLV separation algorithm described above was enabled just prior to burn-out, and computed the optimal separation time to minimize errors in Mach, dynamic pressure, and flight path angle. Separation was commanded at 93.46 seconds from drop, and the Research Vehicle separated successfully and went on to conduct a successful scramjet engine firing⁵. Overall performance dispersions for the launch vehicle are presented in Table 2 compared to pre-flight targets. All dispersions were well within 1σ .

C. Flight 3 Design Changes

The Mach 10 designs for vehicle systems were in place prior to F2, with the exception of the final autopilot and trajectory designs. With only six months planned between flights, limited time was available to analyze the F2 data and incorporate any lessons learned into the F3 design. One major F2 result that was carried over to the F3 design was the limit on transonic alpha. The stack weight was reduced by eliminating ballast, and changing B/AM and aft skirt structures to aluminum from steel. This resulted in a significant aft shift in the stack CG, reducing vehicle static stability substantially. This substantially reduced autopilot margins and challenged the GNC design team to generate a viable gain set for the flight. Limiting maximum alpha to F2 levels improved the margins, however this came at the expense of vehicle performance due to increased drag losses. The mass increase on the FAS and unrealized weight savings on the X-43A and booster adapter, combined with the revised trajectory limits, reduced the nominal separation Mach number below Mach 10 (approximately Mach 9.6). NASA determined that this would not increase risk for the scramjet experiment and elected to proceed with the lower risk

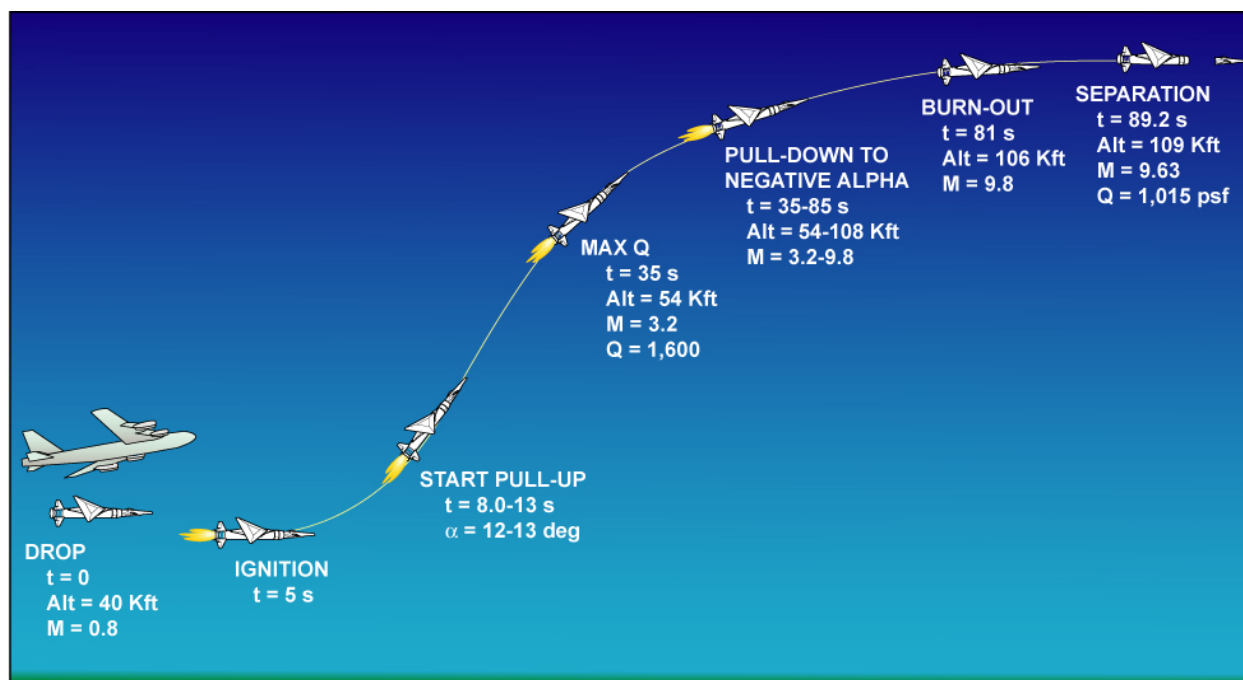


Figure 11. Flight 3 (Mach 10) Nominal Trajectory.

fin leading edges (LEs), shock impingement effects on Silicone Joint, and fin interference regions along the aft skirt and fin root, as shown in Fig. 12. The flow around the fin and fin shaft produces a complex flow interaction condition that augments the heating on the underside of the fin, the shaft housing, and the aft skirt. This was an area of extensive study that is beyond the scope of this paper.

Figure 13 shows a CFD result at a point along the Mach 10 trajectory illustrating shock interaction on the HXLV wing. The shock impingement region is clearly evident and calculated to occur at about the two-thirds position on the wing. An augmentation factor is applied to the heat transfer in this region which is a function of the pressure ratio across the shock. The stronger the shock, the greater the pressure ratio and the greater the augmentation factor. The wing, fins and wing fillet are constructed of graphite epoxy, with a glass transition temperature that dictated a TPS concept consisting of a uniform layer (i.e., constant thickness) of insulator placed over the graphite epoxy. The goal of the insulator is to prevent the graphite epoxy structure from exceeding 90 °F (32 °C) for the duration of the flight, with localized exceedances of this value allowed based on structural loading in that region. Due to much higher stagnation heat loads at the leading edges and additional thickness required as a result, the leading edge tiles were machined with facets that limited the thickness at the transition to acreage TPS. This had a beneficial effect on

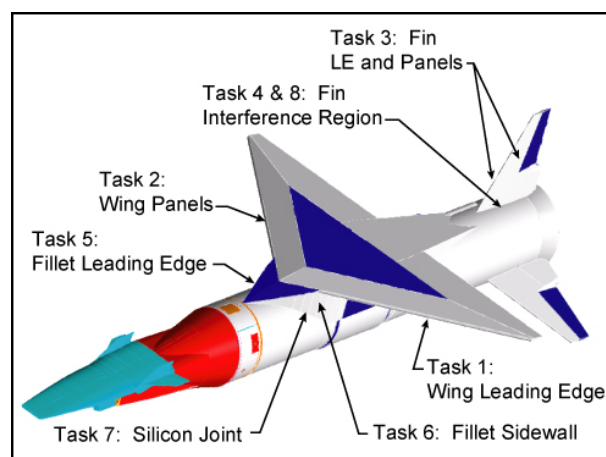


Figure 12. Vehicle Regions for TPS Design Tasks on Flight 3.

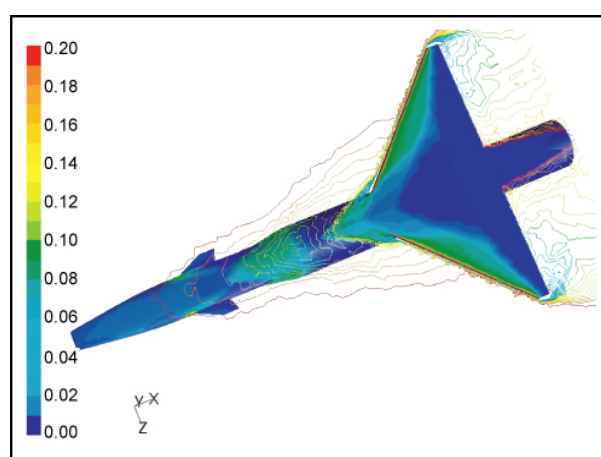


Figure 13. Shock Impingement on Wing at Mach 6.

vehicle drag, which was a concern for the performance-limited Mach 10 configuration. The wing leading edges were constructed from molded carbon phenolic tiles bonded to the wing (Figs. 14 and 15).

D. Flight 3 Results

The X-43A third flight was conducted on 16 November, 2004 originating at the Dryden Flight Research Center, Edwards, CA. Launch day weather was benign, and the B-52 achieved the required drop conditions at Mach 0.8 and 40,000 feet altitude. Overall, the trajectory performance was nominal. Separation conditions for Mach, dynamic pressure, and flight path angle were again within 1σ of preflight predictions. The Mach vs. time plot is shown in Fig. 16, showing a maximum Mach of over 10.0 prior to coast and separation at approximately Mach 9.7, based on balloon sonde data collected just prior to launch. Motor burn was relatively short, and consequently the payload separation algorithm computed and executed an early separation time. Separation attitudes and rates were also well within preflight predictions. Table 3 shows vehicle performance on F3.

Drop transient dynamics were very similar to F2, with a strong right wing down and slight pitch down moment imparted to the stack after hook release for approximately 0.5 – 0.75 seconds. This was likely due to known B-52 local Flowfield effects on the stack, and again the HXLV flight controls captured vehicle rates and attitudes nominally. Figure 17 shows roll rate throughout flight. Again, a modest transonic bump is present, similar to F2 and Pegasus. Figure 18 shows right elevon command versus pre-flight predictions, illustrating the excellent accuracy of the vehicle aerodynamic model for trim de.

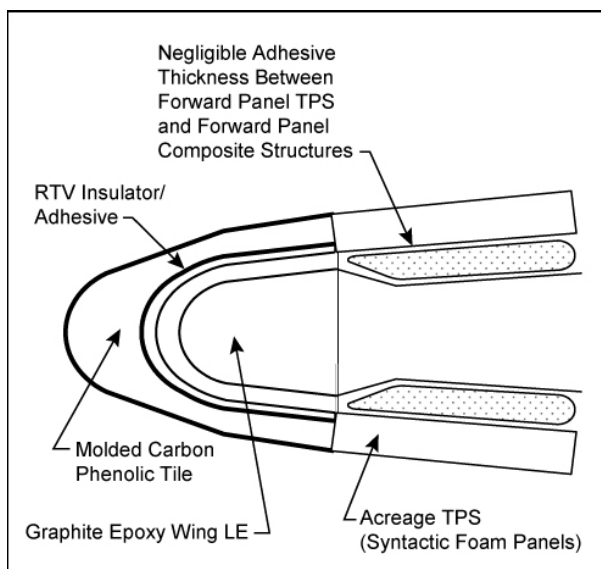


Figure 14. Wing Leading Edge TPS Configuration.

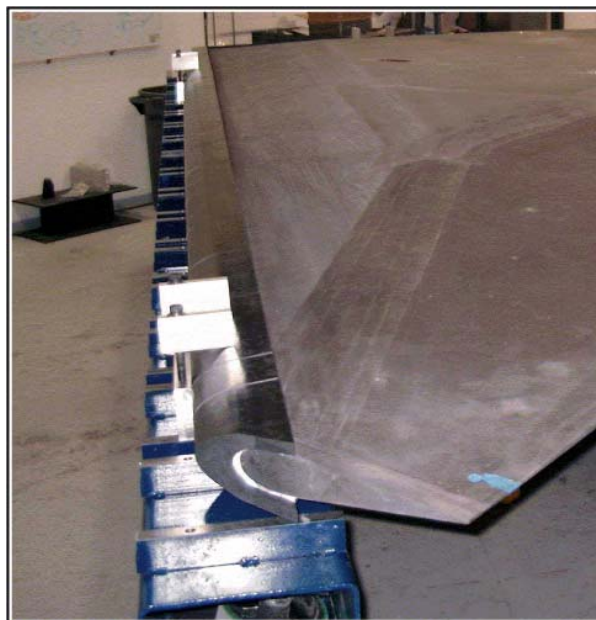


Figure 15. Wing Leading Edge Installation.

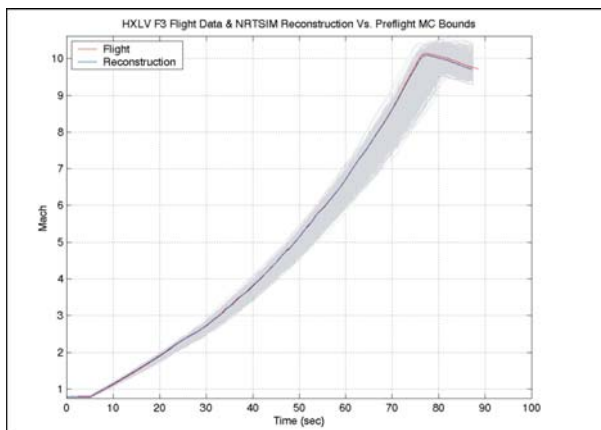


Figure 16. F3 Mach vs. Time.

Table 3. Flight 3 Separation Condition Performance.

Parameter	Target	Flight Data
Separation Time	89.8 sec	88.16 sec
Separation Alt	109,580 ft	109,440 ft
Mach	9.60	9.736
Flight path Angle	1.5 deg	1.69 deg
Dynamic Pressure	1,000 psf	959 psf
Alpha	0.0 deg	0.08 deg
Beta	0.0 deg	-0.13 deg

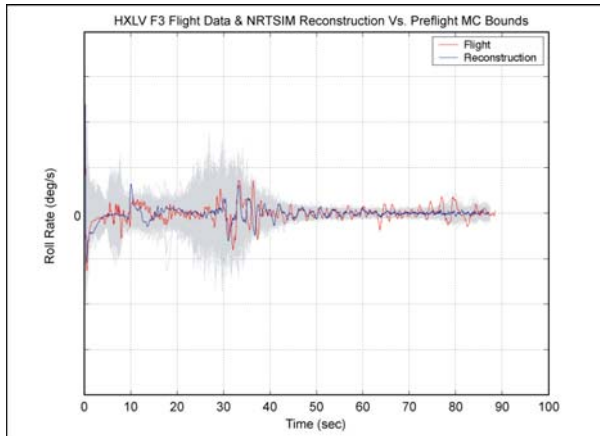


Figure 17. F3 Roll Rate vs. Time.

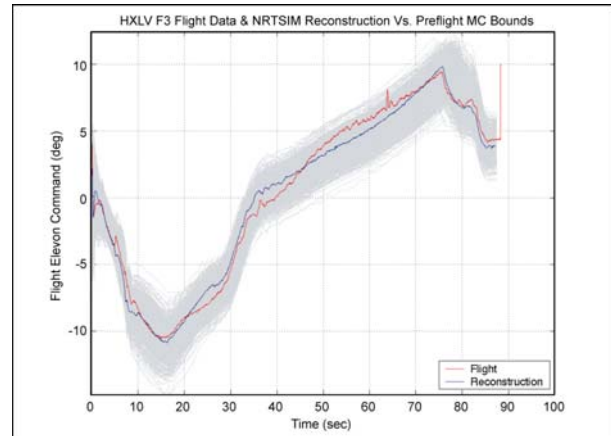


Figure 18. Elevon Deflection vs. Time.

An abrupt disturbance at approximately 64 seconds (Mach 7.4, 92 kft) is apparent in the vehicle dynamics. This disturbance was consistent with other telemetry data that indicate an anomalous shock event at this time. Strain and temperature data from the Wing Fillet indicate a correlated structural shift at this time, indicating a potential yield condition or other localized structural event. Rudder trim was affected by this event, however the vehicle continued to fly nominally until separation. Funding was not available for an in depth analysis of the F3 data, however, and the source of this disturbance remains unknown.

E. LV Path steering Guidance and RV Separation Logic Algorithm

From the beginning of the X-43A program it was clear that the existing Pegasus Stage 1 autopilot architecture was not capable of satisfying the precise trajectory and attitude accuracy requirements specified at the LV/RV separation point. In order to meet the needed dispersion tolerances two major architectural changes to the baseline Pegasus autopilot were made.

The first change was to add a proportional plus derivative type vertical channel path steering guidance law during boosted flight. This law acted as a low bandwidth outer feedback loop around the normal Pegasus pitch channel controller. Output from the guidance law was a pitch angle correction that was added to the open loop pitch angle steering table stored in flight computer memory. This form of guidance significantly improved both flight path angle and dynamic pressure dispersions along the trajectory as the vehicle approaches the separation event. Figure 19 depicts a block diagram representation of the guidance law where:

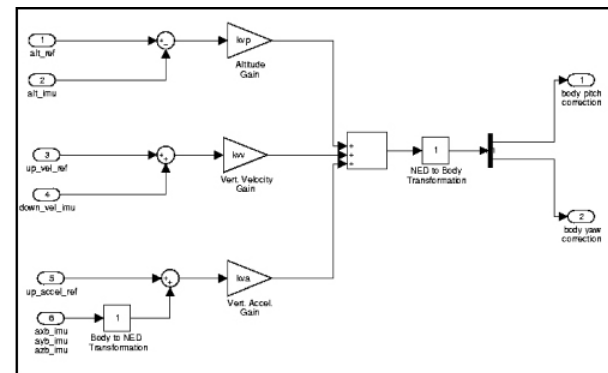


Figure 19. HXLV Path Steering Guidance Law.

alt_ref	=	reference trajectory altitude stored in FC memory
alt_imu	=	IMU measured altitude
up_vel_ref	=	reference trajectory vertical velocity stored in FC memory
down_vel_imu	=	IMU measured down velocity
up_accel_ref	=	reference trajectory vertical acceleration stored in FC memory
axb_imu	=	IMU x-axis measured acceleration
ayb_imu	=	IMU y-axis measured acceleration
azb_imu	=	IMU z-axis measured acceleration
NED	=	north-east-down coordinate frame
kvp	=	gain on altitude error signal
kvv	=	gain on vertical velocity error signal
kva	=	gain on vertical acceleration error signal

The second change made to further improve trajectory state errors at separation was the addition of the RV separation logic algorithm. This algorithm measured vehicle performance at the end of boost to determine the optimum time at which to initiate LV/RV separation. The scheme employed used a simple weighted average (i.e. cost function) of errors in Mach, dynamic pressure, and flight path angle to determine the required time adjustment needed to reach a minimized set of errors in all three states. Equation 3 illustrates the formulation used.

$$t_{\text{sep}} = t_o + W_{q_{\text{bar}}} * [q_{\text{bar}}(t) - q_{\infty o}] + W_{\gamma} * [\gamma(t) - \gamma_o] + W_M * [M(t) - M_o] \quad (3)$$

where: t_{sep} = computed separation time (sec)
 t_o = preflight predicted nominal separation time (sec)
 t = HXLV sequencer time
 q_{bar} = dynamic pressure (psf)
 γ = flight path angle (deg)
 M = Mach number
 o = indicates nominal target separation condition
 W_i = weighting coefficient

Since determination of Mach and dynamic pressure were not directly measurable, they were estimated on board using GRAM95 atmospheric model data stored in flight computer memory in combination with IMU navigation data. The weighting coefficients were developed based on the separation dispersion requirements defined for each mission. In the event a significant trajectory error existed at the end of motor burn (i.e. outside preflight predictions) the algorithm also employed a predetermined time-out to initiate RV separation with the intent that separation would be guaranteed to occur, even if it was not at the optimal time, in an attempt to salvage some portion of the RV flight test experiment.

V. General Considerations and Lessons Learned

The second and third flights of the HXLV validated the concept of utilizing a Pegasus derivative for Hypersonic flight testing up to Mach 10. The accuracy of the separation conditions at Mach 7 and 10 was extremely high given the inherent total impulse uncertainties associated with solid rocket motors, atmospheric variations, winds, and limited flight data on the HXLV airframe. This accuracy was achieved due to several factors as outlined below.

The first factor was the fidelity of the models developed as part of the baseline design (INS, structural, thermal, motor ballistics) and those developed as part of the RTF efforts (FAS non-linear model, aerodynamic models). In addition to the large investment made in the primary models, NASA invested in either verification activities conducted by independent experts, validation testing, or in many cases both. For example, in addition to the seven wind tunnel entries conducted following the first flight to populate a sparse wind tunnel data matrix, NASA conducted full Navier-Stokes CFD studies of vehicle aerodynamics, including detailed modeling of vehicle structures to aid in Aeroelastic structural analysis. Roll disturbance models were created and iteratively applied in varying degrees of severity in an attempt to “break” the autopilot and cause loss of vehicle control. Rotary derivatives, notoriously difficult to obtain in wind tunnel testing, were parametrically varied to determine the impacts of different derivations from flight data. In addition, extensive validation testing of the redesigned FAS validated FAS non-linear models throughout, and substantially outside, the HXLV anticipated flight envelope. This testing was conducted at significant time and expense, however the requirement to minimize flight risk on the second and third flights demanded this approach. Thermal models and TPS designs were developed by first taking Monte Carlo results, identifying the worst-case Design Reference Mission (DRM), and then placing safety factors on top of the analyses. This approach resulted in a robust design for the composite Pegasus structures, which would not survive any significant heating or local failure of TPS. Where uncertainties existed, for example in the fin interference region between fin root and aft skirt, additional modeling and verification tasks were undertaken. Significant safety factors were included in all designs, rendering the actual flights quite benign compared with pre-flight Monte Carlo analysis and qualification testing.

The second factor was continuity of the design team. Although a failure of the magnitude of the F1 boost failure can often lead to significant personnel changes on a program, the Orbital design team remained intact with few exceptions. Thus the findings of the MIB, which was supported extensively by Orbital engineering, were carried over successfully to all aspects of the design. Nuances of the autopilot behavior, airframe responses to disturbances, critical FAS modeling parameters, and complex aerodynamics were all incorporated in to design, verification, and test of the F2 and F3 systems.

The third factor was a willingness to take appropriate risks. The RTF and F2 and F3 integration efforts were not without setbacks and difficult decisions. These included the decision to proceed with a propellant offload that significantly altered the motor ballistics, ignition characteristics, and other design margins without a static firing. The Program made decisions on both F2 and F3 to rely on robust acceptance and pre-launch testing for the FAS to prove flight-worthiness, which had several units fail testing as a result of a marginal design for noise rejection. This was perhaps one of the most difficult decisions program leadership was faced with. Since the FAS was developed quickly as part of the RTF effort, to stop and redesign the system would have taken up to a year and added substantially to the cost of the program. In both flights, the FAS performed flawlessly.

A. Lessons learned

Shrouded vs. unshrouded payloads. Hypersonic flight test programs conducting testing with exposed payloads--either lifting bodies, engines, or other complex shapes--should perform a trade of shrouded versus unshrouded for the overall system. Clearly, a shrouded payload (shroud designs for existing launch vehicles are available in a wide variety of sizes and shapes) provides a lower risk design for the launch vehicle aerodynamic and control system teams. Boosting to exoatmospheric conditions (Q_{bar} below 5 psf), deploying a shroud and then the research payload, and allowing the payload vehicle to reenter and pull up to the desired test condition results in a lower risk launch vehicle development. The penalties for this approach can be great for the test article, however, and include large structural loads, and much larger peak heating loads than a direct insertion to the test condition.

Performance Margins at the start of a Program. During trade studies for the Hyper-X program in late 1996, Orbital estimated approximately 1000 lbm margin for the Mach 10 mission. Several developments over the life of the program eroded that margin until the Mach 10 separation condition was not achievable on the third flight. These can be categorized as follows: vehicle stiffness requirements that significantly increased the RV and adapter weights above the baseline; changes made to the system to improve control systems margins; conservatism in TPS designs that affected weight as well as drag; and imposing trajectory limits based on stability and control concerns derived from the final stack mass properties. Although the Mach 10 separation could have been achieved with substantial additional risk, the program elected to accept the final nominal Mach 9.6 separation condition and still obtained excellent data. Thus, what started as a very generous performance margin (over 2.5 percent in terms of ballast), was completely gone by the third flight. Programs without a high fidelity model of the demonstrator/launch vehicle system should utilize 5 percent of gross stack weight as margin at the start of the program.

Thermal Design Approach for Hypersonic LVs. Thermal design considerations for the HXLV are discussed elsewhere in this paper. Due to the complexity of the flow fields, the severity of the stagnation heating and total gas temperature, and adaptation of heritage launch systems designed for much lower thermal environments, great care must be taken to ensure all areas of the vehicle are completely protected. Areas that require no TPS at all for a standard launch vehicle application can cause structural failures and loss of mission on hypersonic flight tests if not properly protected. Wherever possible, launch vehicles with excess performance should be selected so that a low-cost, high-margin TPS can be designed to minimize risk for the LV.

Control System Approaches for Hypersonic LVs. For vehicles such as the HXLV, care must be taken to ensure large control power margins (>50 percent) for the first flight of the system. As discussed above, thoroughly validated high fidelity models of the control system are mandatory due to the high loading and consequent narrow flight envelopes of these systems. Validation of the models should be conducted at the highest level of system integration possible, ideally with flight-like structural elements in the system or flight hardware.

B. Looking Ahead

The X-43A Program validated the approach of utilizing solid rocket motor based launch vehicles, and Pegasus in particular, for hypersonic flight testing. As illustrated in the booster trades section of this paper, many alternatives are available for this testing. Other experimenters have utilized converted surface to air missiles and sounding rockets for captive scramjet flight tests. However, for airframe-integrated testing or where precise test conditions are required, a guided system is the optimal choice. Well-characterized path-steering guidance strategies and separation condition optimization algorithms can be implemented on guided launch vehicles, as they were so successfully on X-43A.

Review of the current inventory of operational guided suborbital and space launch vehicles reveals a broad range of capability. These include single and multi-stage variants of Pegasus (Orion), Minotaur, Minuteman, Peacekeeper, and Castor IVB. These systems, with modifications comparable to the HXLV or less significant, could provide flight test options for a wide variety of hypersonic test vehicles or technologies, from Machs 5 – 20. An example of a recoverable technology demonstrator on an HXLV is shown in Fig. 20. This configuration was studied for a Scramjet engine flight test at Mach 8 with payload recovery.

For the X-43C Program, Orbital studied conversion of the HXLV to carry the heavier and larger X-43C Demonstrator Vehicle to a Mach 6 Flight Test Condition in some detail prior to that program's termination in December, 2003. The notional X-43C configuration is shown in Fig. 21. Using the existing X-43A vehicle systems and the standard Pegasus interface with Orbital's L-1011 Stargazer Carrier Aircraft, Orbital evaluated the mission requirements for the X-43C program and determined that the Mach 7 variant (with offload) of the X-43A HXLV would meet X-43C requirements. The L-1011 provides a flexible launch platform for X-43C and other hypersonic missions, with a heavy lift capability up to 51,000 lbm and greater flexibility on CG location than other platforms. For X-43C, we conducted performance trade studies of a number of possible launch vehicle configurations, including different propellant offloads, ballast configurations, deletion of the B/AM, and varying drop altitudes.

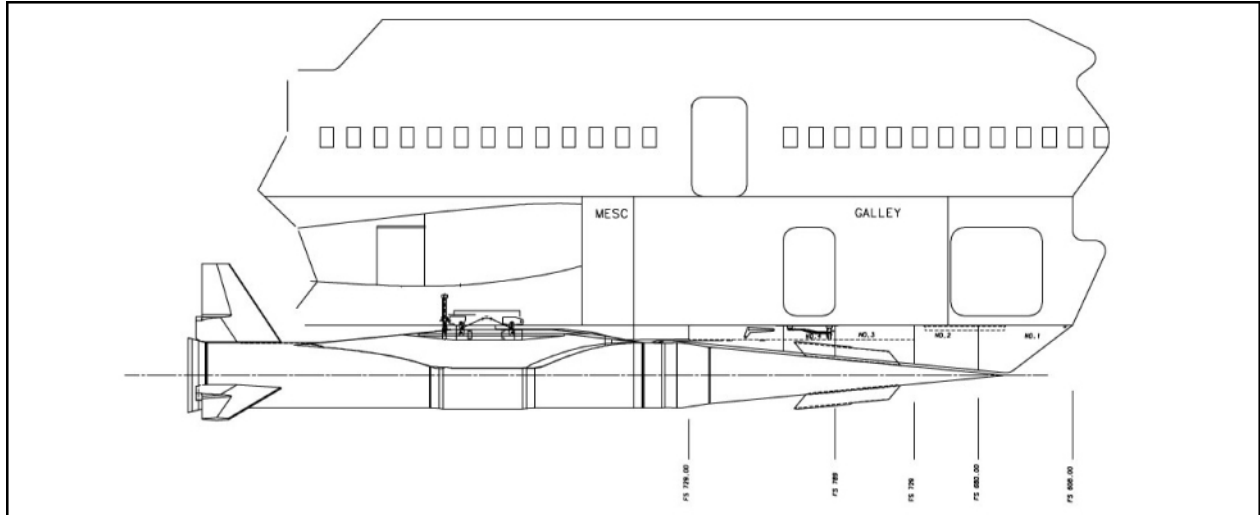


Figure 20. Recoverable Scramjet Experiment on HXLV Integrated on the L-1011.

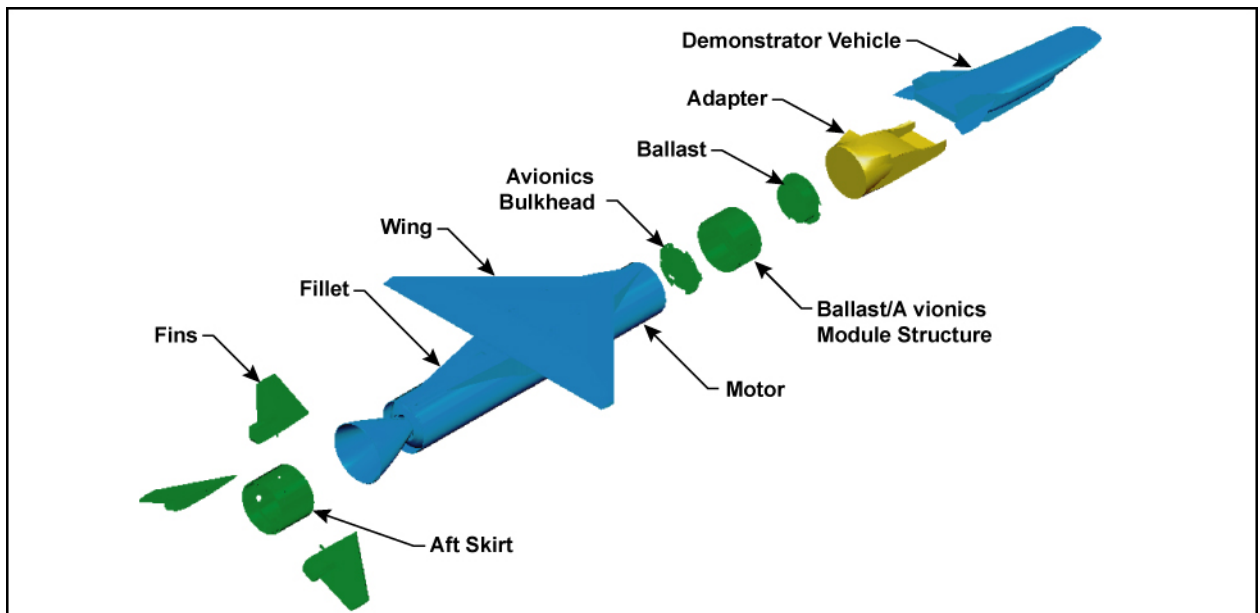


Figure 21. X-43C HXLV Configuration.

VI. Conclusions

The HXLV provided unprecedented test condition accuracy for the second Mach 7 and Mach 10 flight tests on the X-43A program. Through the F1 failure, MIB, and RTF activities prior to F2, many valuable lessons were learned regarding Hypersonic launch vehicle design that are included in this paper. The challenges presented by the unique airframe and aerodynamic mold line of the HXLV, combined with the unforgiving flight environments characteristic of hypersonic flight tests, can be addressed and overcome with highly refined designs. Future programs considering hypersonic flight testing should adapt lessons learned on Hyper-X at the start of their endeavor to ensure a well-scoped program that will yield a successful data collection.

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